

Gesture Sensor

Objective:

To detect hand gestures such as Left, Right, Up, and Down using the built-in APDS9960 gesture sensor on the Arduino Nano 33 BLE Sense.

Procedure:

1. Opened Arduino IDE.
2. Installed the Arduino_APDS9960 library.
3. Opened the GestureSensor example from:
4. File → Examples → Arduino_APDS9960 → GestureSensor
5. Uploaded the program to the Arduino Nano 33 BLE Sense board.
6. Opened the Serial Monitor at 9600 baud rate.
7. Moved my hand in different directions near the sensor.
8. Observed the detected gestures displayed in the Serial Monitor.

Working Principle:

The APDS9960 contains:

- Infrared (IR) LED
- Multiple IR detectors

The sensor emits infrared light and measures the reflected light from a moving hand.

By analyzing the sequence of reflected signals, it determines the direction of movement:

- Left Gesture
- Right Gesture
- Up Gesture
- Down Gesture

Observations:

Message (Enter to send message to Arduino Nano 33 BL)

Detecting gestures ...

Detected RIGHT gesture

Detected RIGHT gesture

Detected LEFT gesture

Detected DOWN gesture
